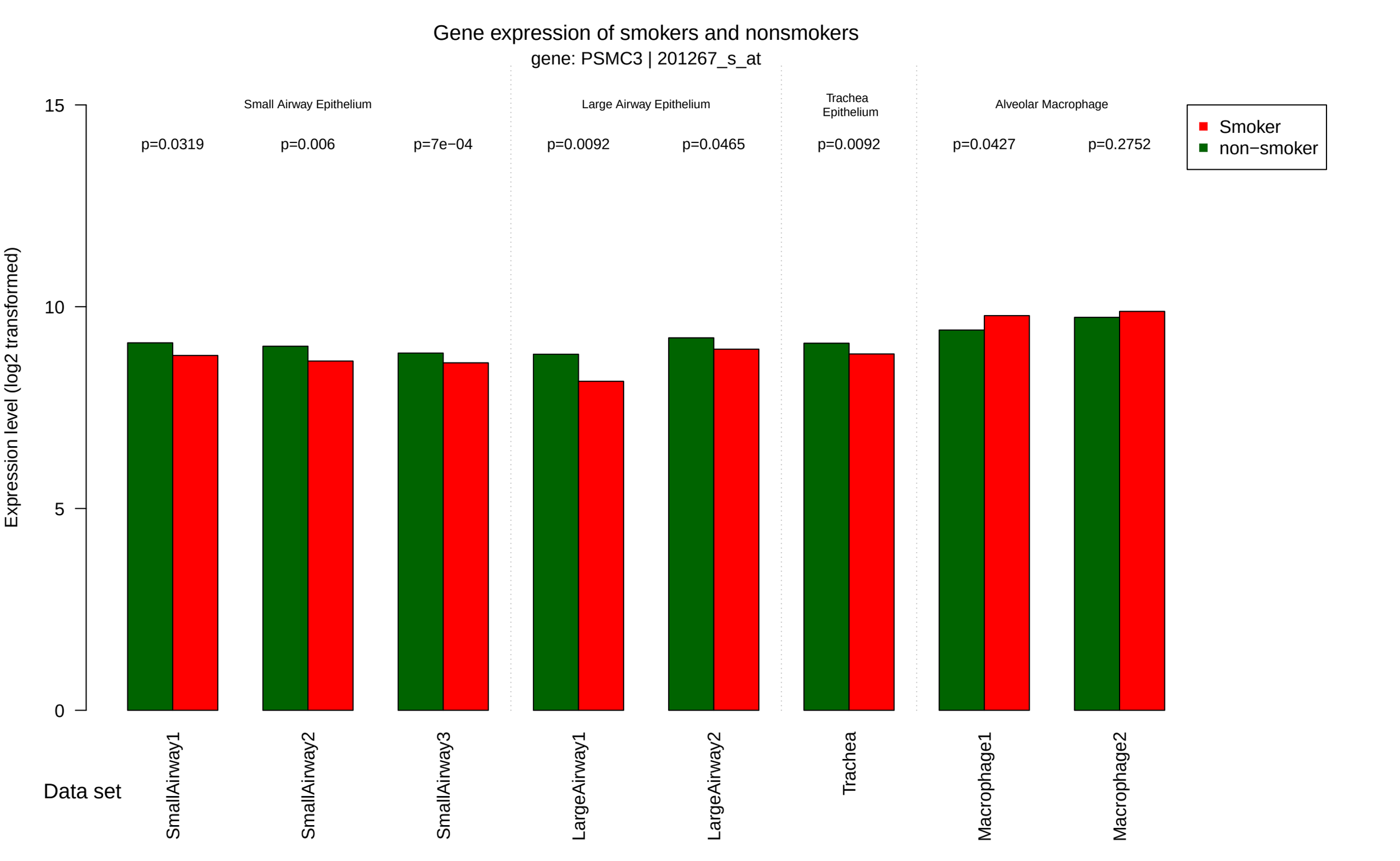
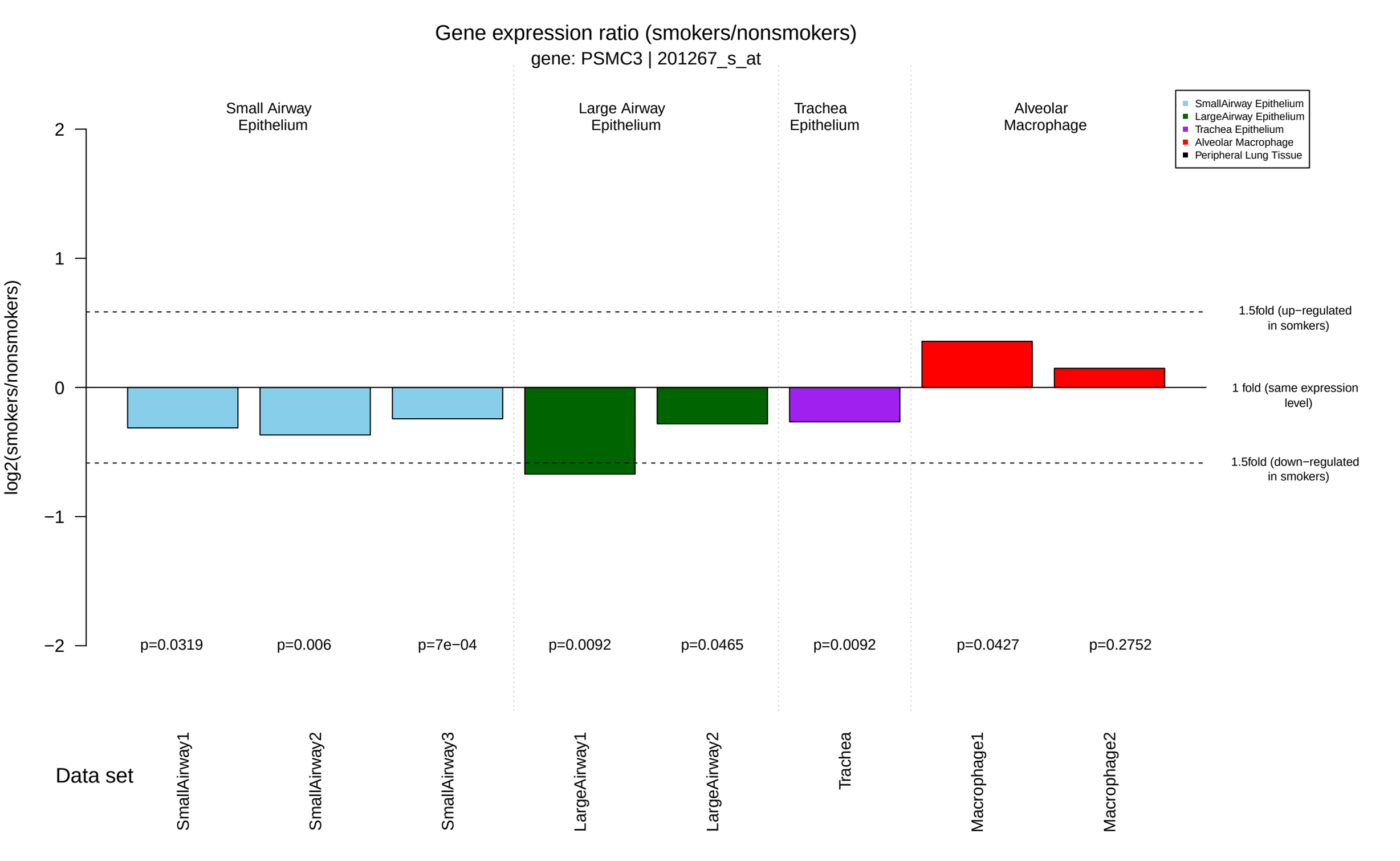
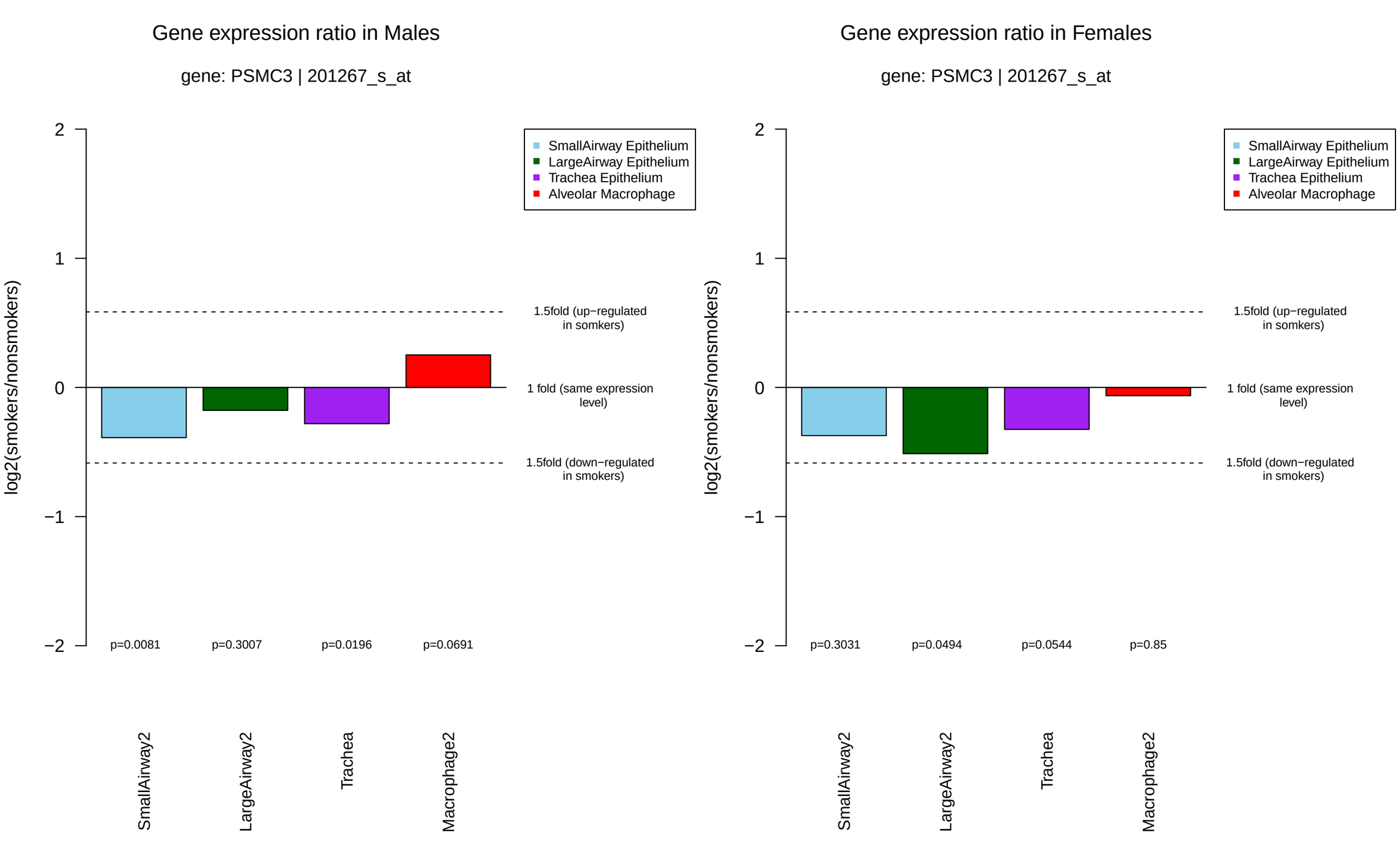
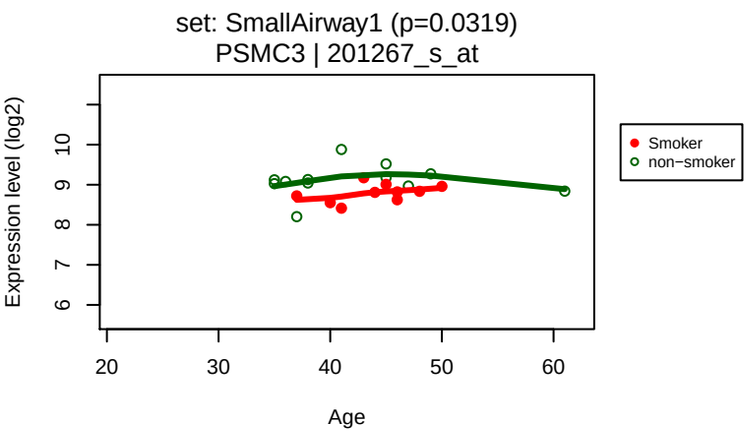
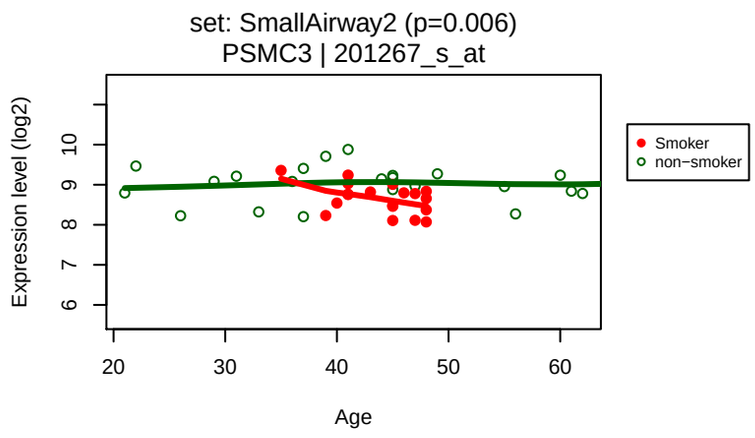
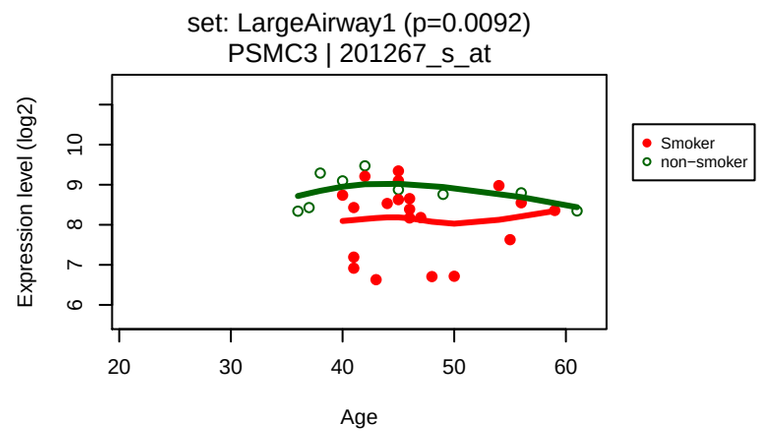

Gene expression of smokers and nonsmokers
gene: PSMC3 | 201267_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0319
p=0.006
p=7e−04
p=0.0092
p=0.0465
p=0.0092
p=0.0427
p=0.2752
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

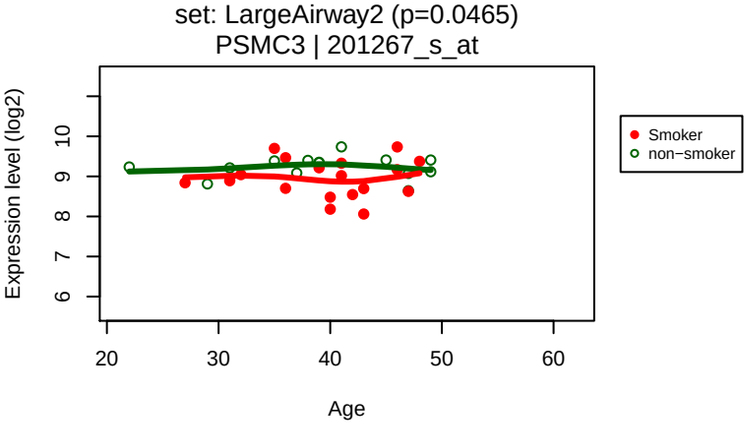

Gene expression ratio (smokers/nonsmokers)
gene: PSMC3 | 201267_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0319
p=0.006
p=7e−04
p=0.0092
p=0.0465
p=0.0092
p=0.0427
p=0.2752
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

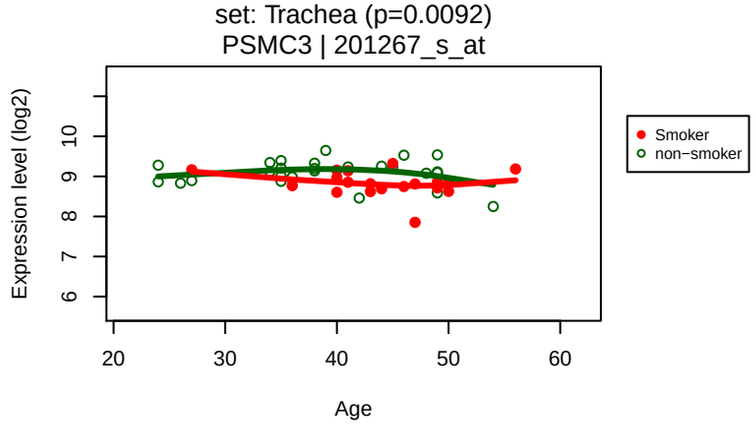

Gene expression ratio in Males
gene: PSMC3 | 201267_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0081
p=0.3007
p=0.0196
p=0.0691
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMC3 | 201267_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3031
p=0.0494
p=0.0544
p=0.85
SmallAirway2
LargeAirway2
Trachea
Macrophage2

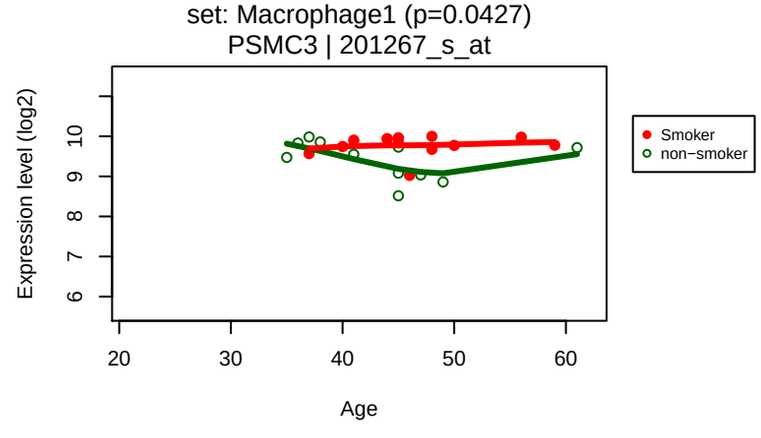

set: SmallAirway1 (p=0.0319)
PSMC3 | 201267_s_at
Expression level (log2)
Age
Smoker
non−smoker

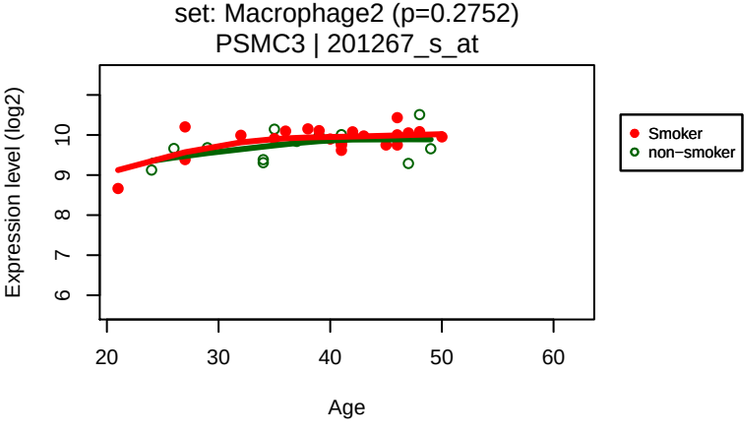

set: SmallAirway2 (p=0.006)
PSMC3 | 201267_s_at
Expression level (log2)
Age
Smoker
non−smoker

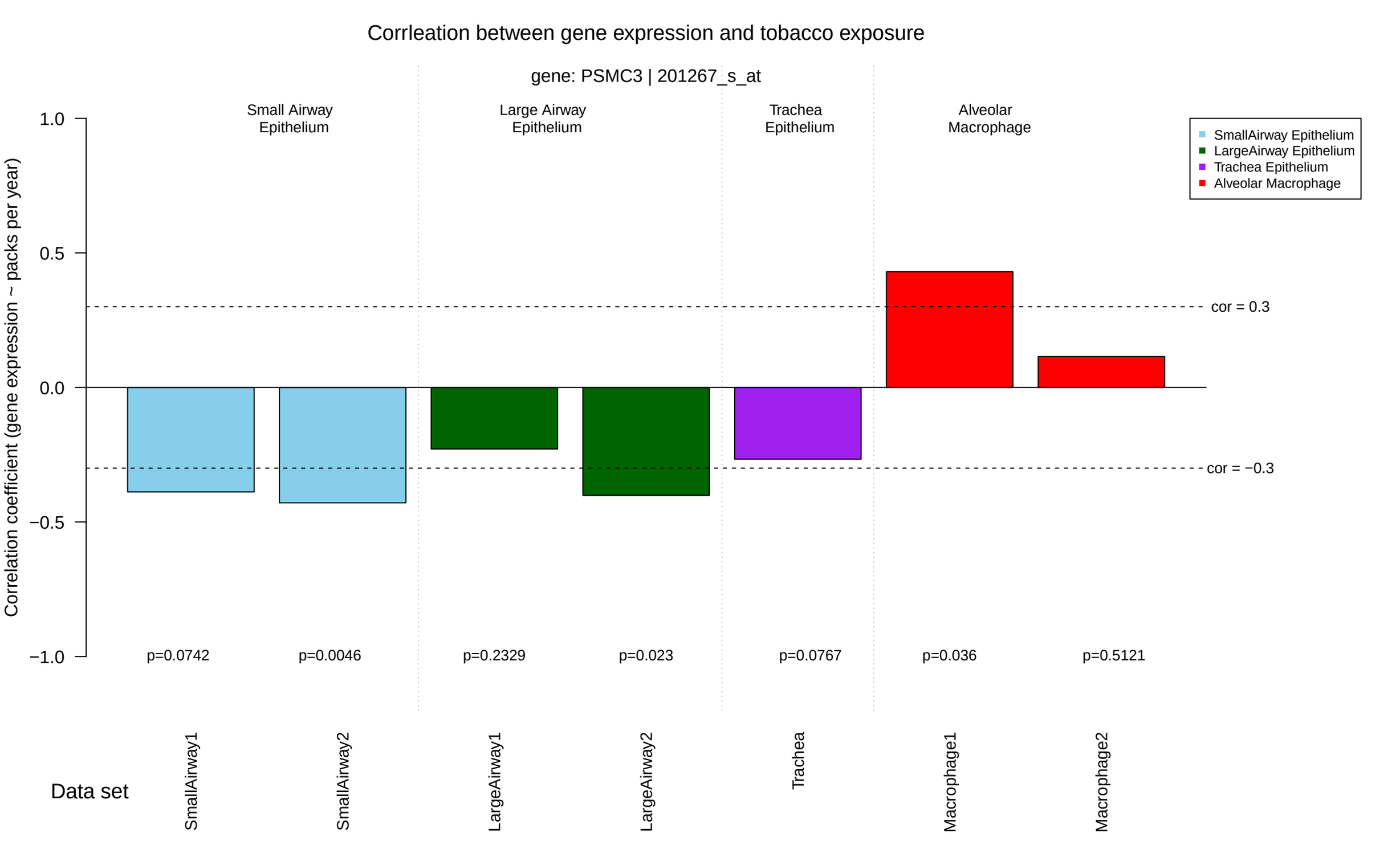

set: LargeAirway1 (p=0.0092)
PSMC3 | 201267_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0465)
PSMC3 | 201267_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0092)
PSMC3 | 201267_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0427)
PSMC3 | 201267_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2752)
PSMC3 | 201267_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMC3 | 201267_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0742
p=0.0046
p=0.2329
p=0.023
p=0.0767
p=0.036
p=0.5121
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2